HERITAGE SHIELD

Domain: Simulation and Digital Twin
Category: Software

Team ID: 031
Team Name: Qualified

Problem Statement

Predictive Conservation of India's Cultural Heritage through AI-Assisted Digital Twins

- Heritage sites are continuously exposed to structural deterioration, environmental stress, human activity and natural hazards. [2][3]
- However, condition assessment and conservation planning often rely on fragmented records and periodic inspections, making it difficult to detect deterioration trends early and prioritize preventive intervention before damage becomes irreversible. [2][3][5]
- Affected Stakeholders:
  - ASI / State heritage authorities
  - Conservation architects & structural experts
  - Site managers / field officers
  - Local authorities responsible for heritage assets
  - Conservation teams working with limited inspection resources
- This needs to be fixed because late intervention is expensive and sometimes impossible. The conservation of culturual sites is non-negotiable for the conservation of India’s cultural identity.

Literature Survey & Existing Solutions

Approach	What it provides	Limitation / gap
Manual visual inspection	Expert assessment of visible condition	Periodic, subjective and difficult to compare [3][4]
3D laser scanning / LiDAR	High-resolution geometry [5]	Primarily geometric documentation; acquisition can be costly.
Photogrammetry + Digital Twin	High-fidelity 3D representation and deterioration monitoring [3]	Demonstrated in research, but generally case-specific [3][5]
AI visual inspection	Automated detection/assistance [4]	Detection ≠ longitudinal asset management or intervention prioritisation
Heritage Digital Twin + IoT/ML	Links digital representation with monitoring/prediction [2] [5]	Emerging research direction; practical scalable integration remains an open challenge [5]

The Government is already digitising heritage. We add to that by not just keeping records, but analysing them to maintain our national identity. [1][2]

Proposed Idea

Identified Gap: Existing approaches address individual stages such as documentation, inspection, or checking. A practical implementation gap remains in connecting repeated observations to the same heritage components, tracking deterioration, incorporating risk context and converting this evidence into conservation priorities. [2][3][4][5][6]

Heritage Shield is an AI-assisted digital-twin platform that converts fragmented and periodic heritage-condition observations into a continuously updated record of asset condition, deterioration trends and risk, helping conservation authorities identify which heritage components require attention first.

Observe → **Digital Twin** → **AI Assess** → **Temporal Change**

Risk and Health → **Estimate Deterioration** → **Prioritize** → **Conserve**

From fragmented heritage evidence → to a living, predictive conservation decision system.

Tech stack / Technology

1. Data Input: historical photographs, inspection records, GIS data

2. Digital Twin: Photogrammetry/3D Reconstruction, Blender, COLMAP

3. AI Assessment: Python, OpenCV, ML/CV

4. Risk Intelligence: ML, Risk Scoring

5. Platform: Next.js/React, PostgreSQL, FastAPI, PostGIS

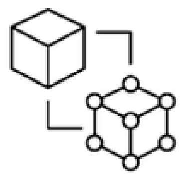
6. Automation Layer: Automated data ingestion → reconstruction → condition assessment → temporal comparison → risk scoring → priority generation

• Photogrammetry → Automated 3D reconstruction •Blender → Model optimisation •Computer Vision → Damage detection •ML → Deterioration & risk prediction •GIS/PostGIS → Spatial & hazard mapping •Three.js → Interactive digital twin •Python/FastAPI → AI & backend processing •PostgreSQL → Heritage data & history •Cloud → Scalable automation •IoT (Future) → Real-time monitoring

Innovation / Differentiation

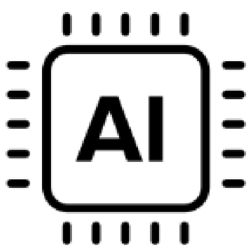
Our differentiation lies in integrating digital reconstruction, component-level condition history, temporal change detection and risk-based prioritisation into one conservation workflow.

Living Digital Twin :



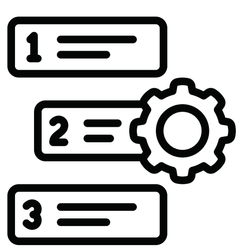
Creates a continuously evolving digital representation of the heritage asset, including component-level mapping and condition history.

Longitudinal AI-Powered Intelligence :



Combines AI-assisted condition assessment, historical–current comparison and temporal change detection to understand how deterioration evolves over time, rather than assessing damage as an isolated event.

Predictive Risk-Based Prioritisation :



Integrates condition, environmental stress and hazard exposure to predict deterioration and rank assets/components according to where conservation attention is needed first.

Feasibility & Expected Impact

1. **Focused and Feasible MVP :**

   Validate the complete Heritage Shield workflow on 1–2 heritage sites using historical photographs, inspection records, geospatial data, environmental information and hazard maps.

2. **Scalable Implementation :**

   The platform can scale progressively from individual sites → city/district level → state-level monitoring → national heritage intelligence network. Can scale by adding IoT sensors, drones, larger historical datasets, predictive models, automatic scan-to-twin.

Expected Impact :

Earlier deterioration detection • Disaster Management • Evidence-based prioritisation • Risk-informed conservation • Better allocation of limited resources

References

[1] **NMMA***

Ministry of Culture, Government of India. Documentation and National Heritage Database.

[2] Jouan & Hallot, 2020*

Digital Twin: Research Framework to Support Preventive Conservation Policies. ISPRS International Journal of Geo-Information, 9(4), 228.

[3] Kong & Hucks, 2023*

Preserving our heritage: A photogrammetry-based digital twin framework for monitoring deteriorations of historic structures. Automation in Construction, 152, 104928.

[4] Mishra & Lourenço, 2024*

Artificial intelligence-assisted visual inspection for cultural heritage: State-of-the-art review. Journal of Cultural Heritage, 66, 536–550.

[5] Colace et al., 2026*

A computational approach to predictive conservation. Digital Applications in Archaeology and Cultural Heritage, 40, e00519.

***links attached**

Team composition

Member no	Skills
Member 1	Python, Backend, API Integration
Member 2	3D Modelling, Photogrammetry & Digital Twins
Member 3	AI/ML, Computer Vision & Damage Detection
Member 4	GIS, Geospatial Data & Risk Mapping
Member 5	Frontend, Three.js & Dashboard Development
Member 6	Data Processing, Testing & System Integration